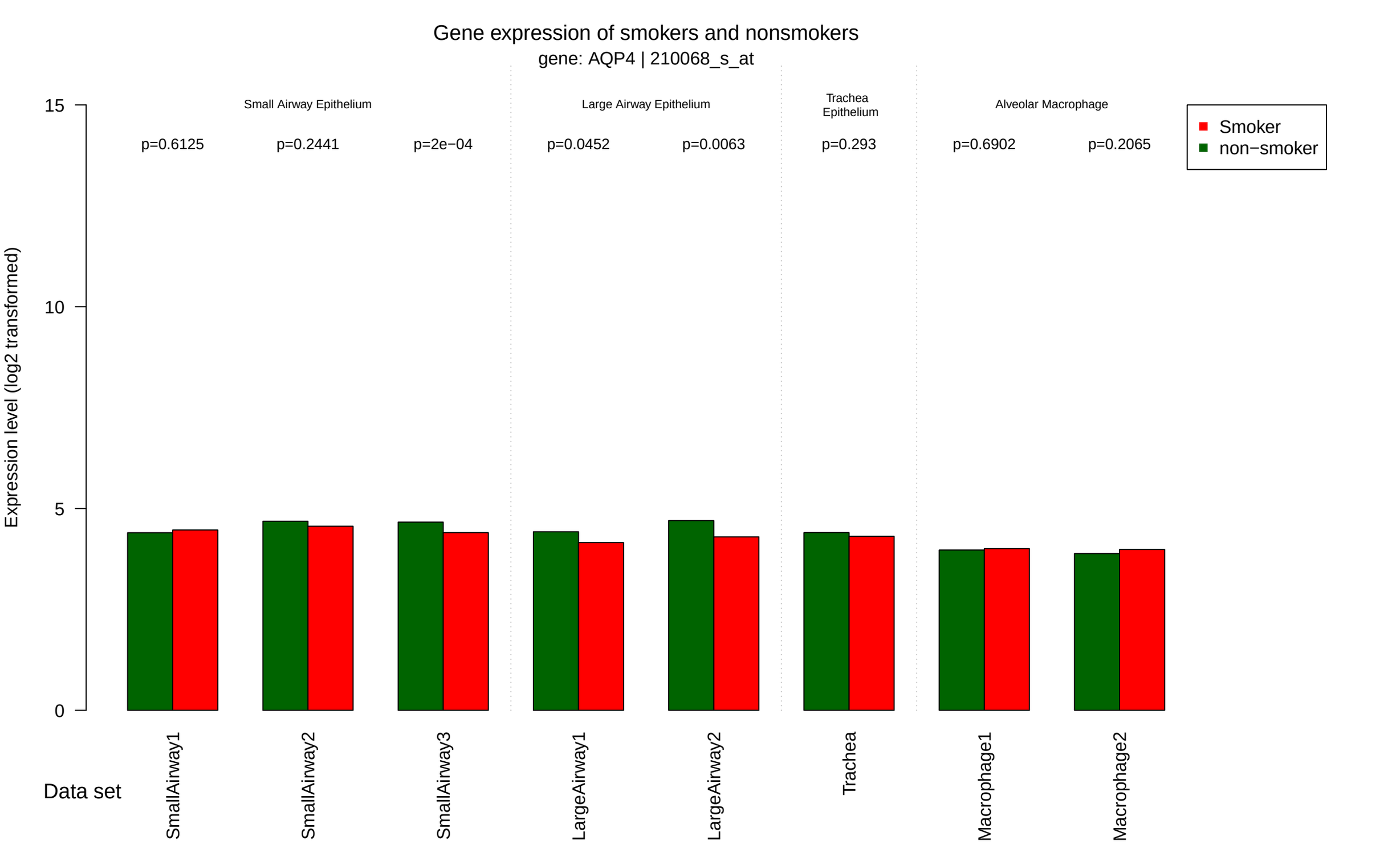
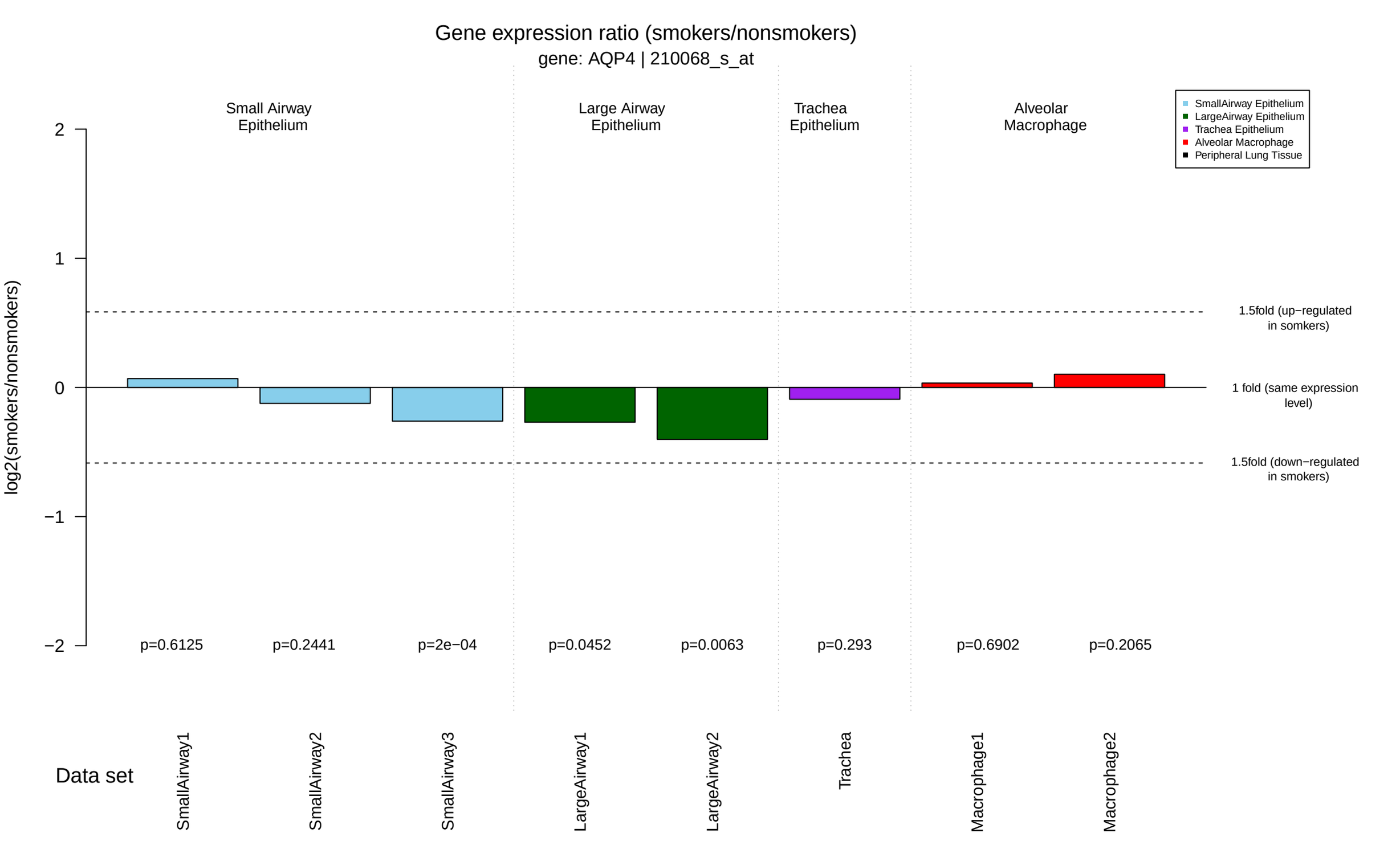
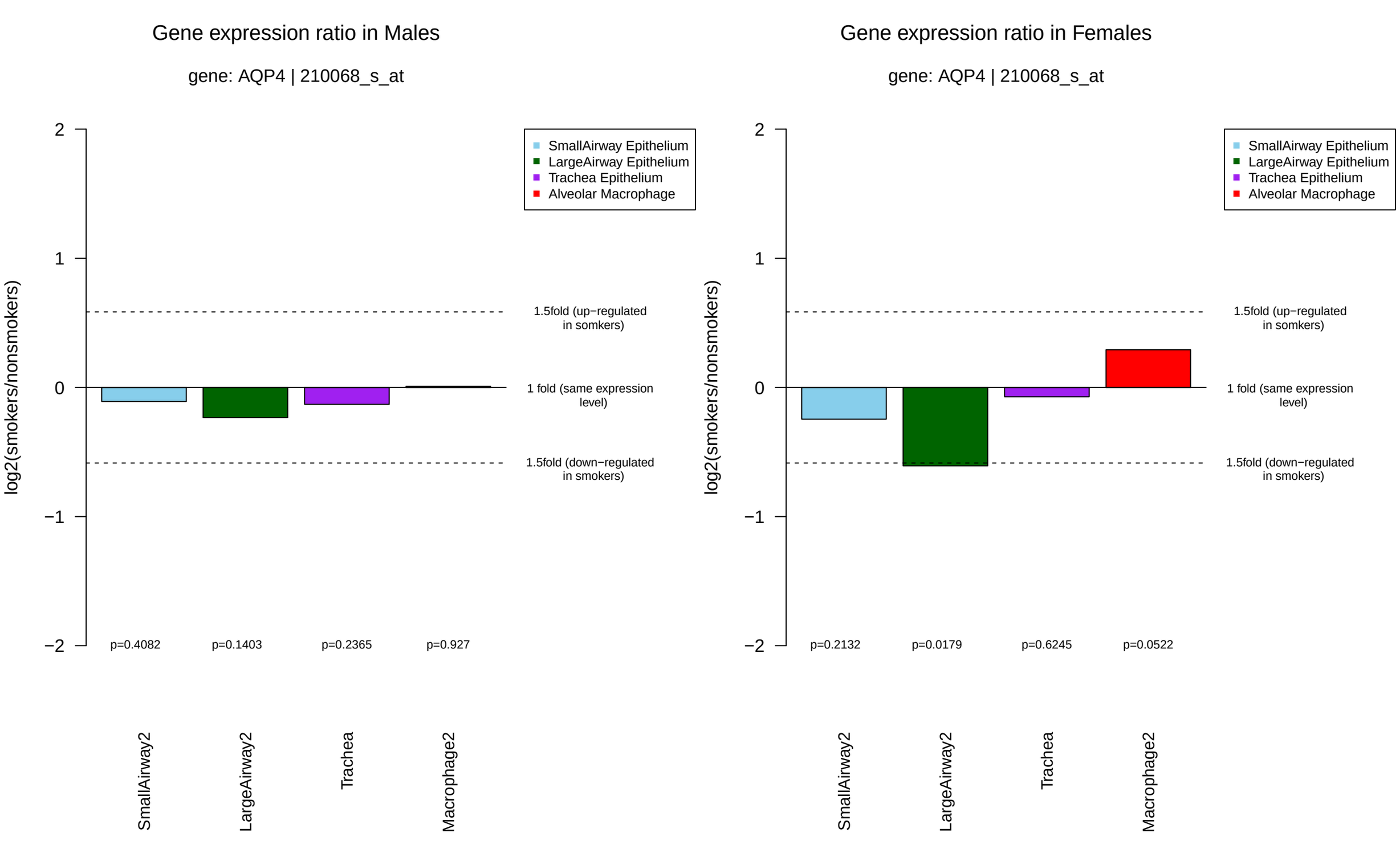
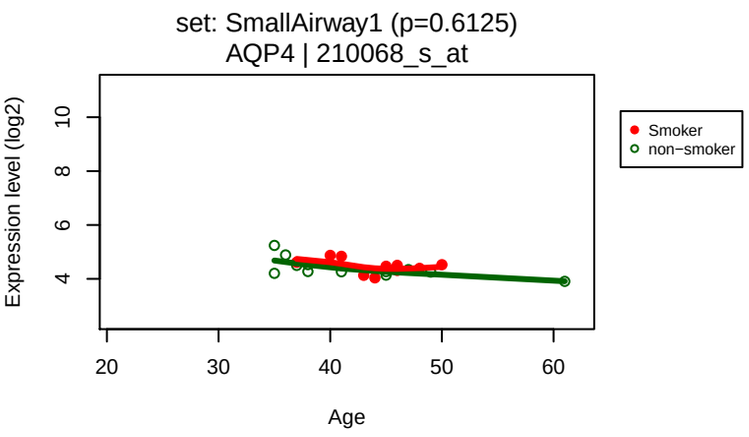
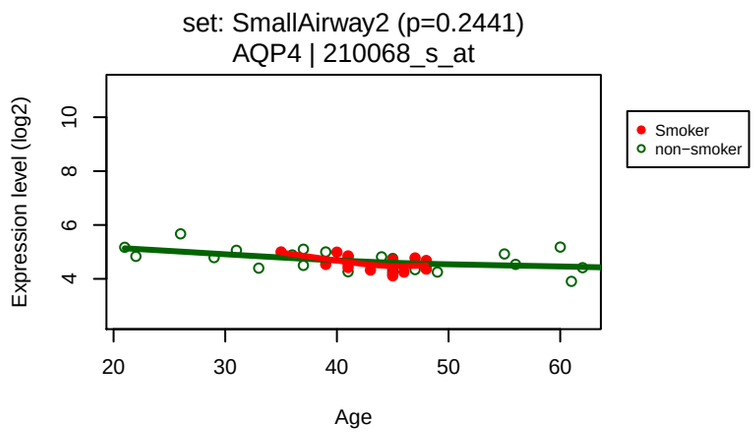
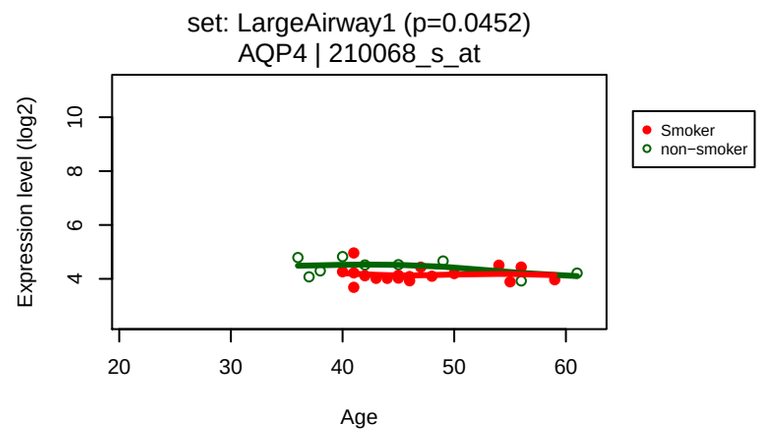

Gene expression of smokers and nonsmokers
gene: AQP4 | 210068_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6125
p=0.2441
p=2e−04
p=0.0452
p=0.0063
p=0.293
p=0.6902
p=0.2065
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

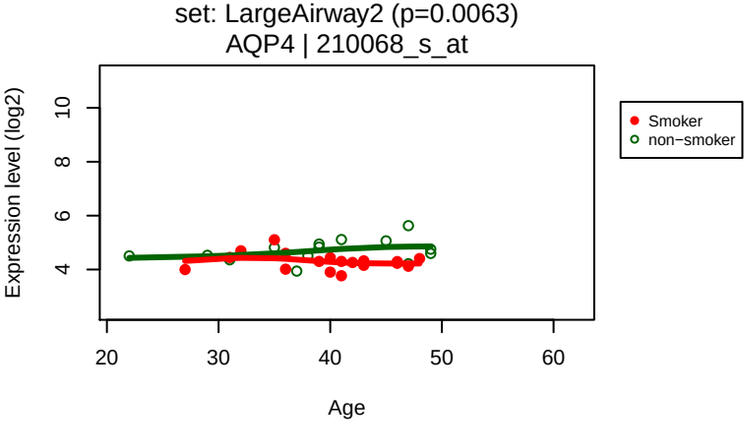

Gene expression ratio (smokers/nonsmokers)
gene: AQP4 | 210068_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6125
p=0.2441
p=2e−04
p=0.0452
p=0.0063
p=0.293
p=0.6902
p=0.2065
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

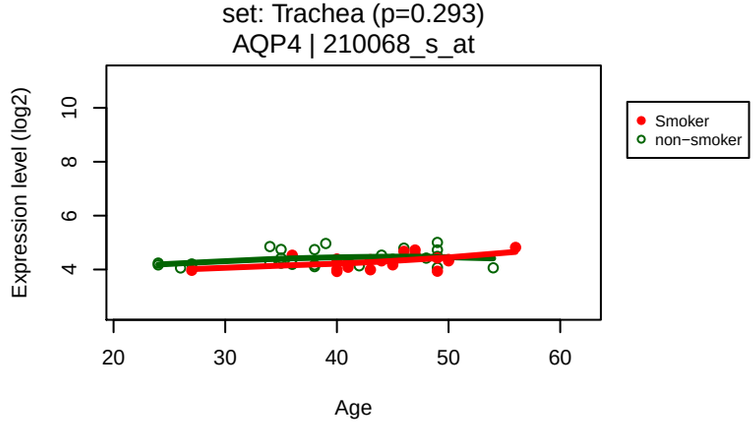

Gene expression ratio in Males
gene: AQP4 | 210068_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4082
p=0.1403
p=0.2365
p=0.927
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP4 | 210068_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2132
p=0.0179
p=0.6245
p=0.0522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

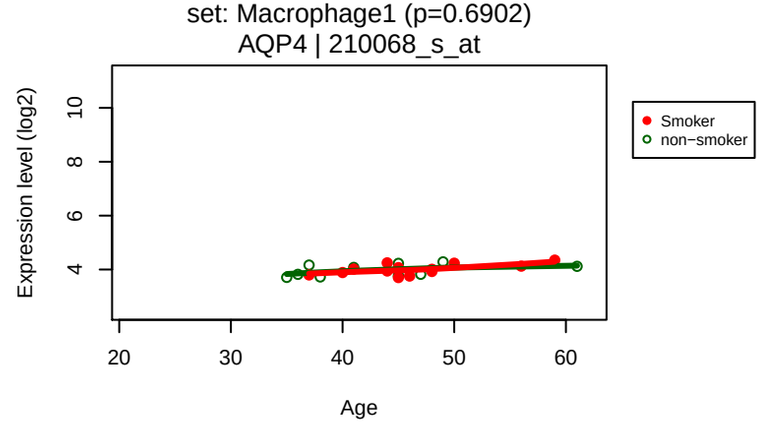

set: SmallAirway1 (p=0.6125)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker

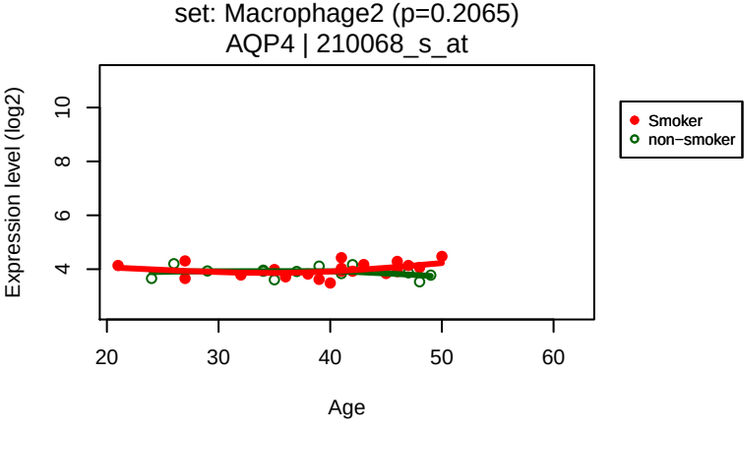

set: SmallAirway2 (p=0.2441)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker

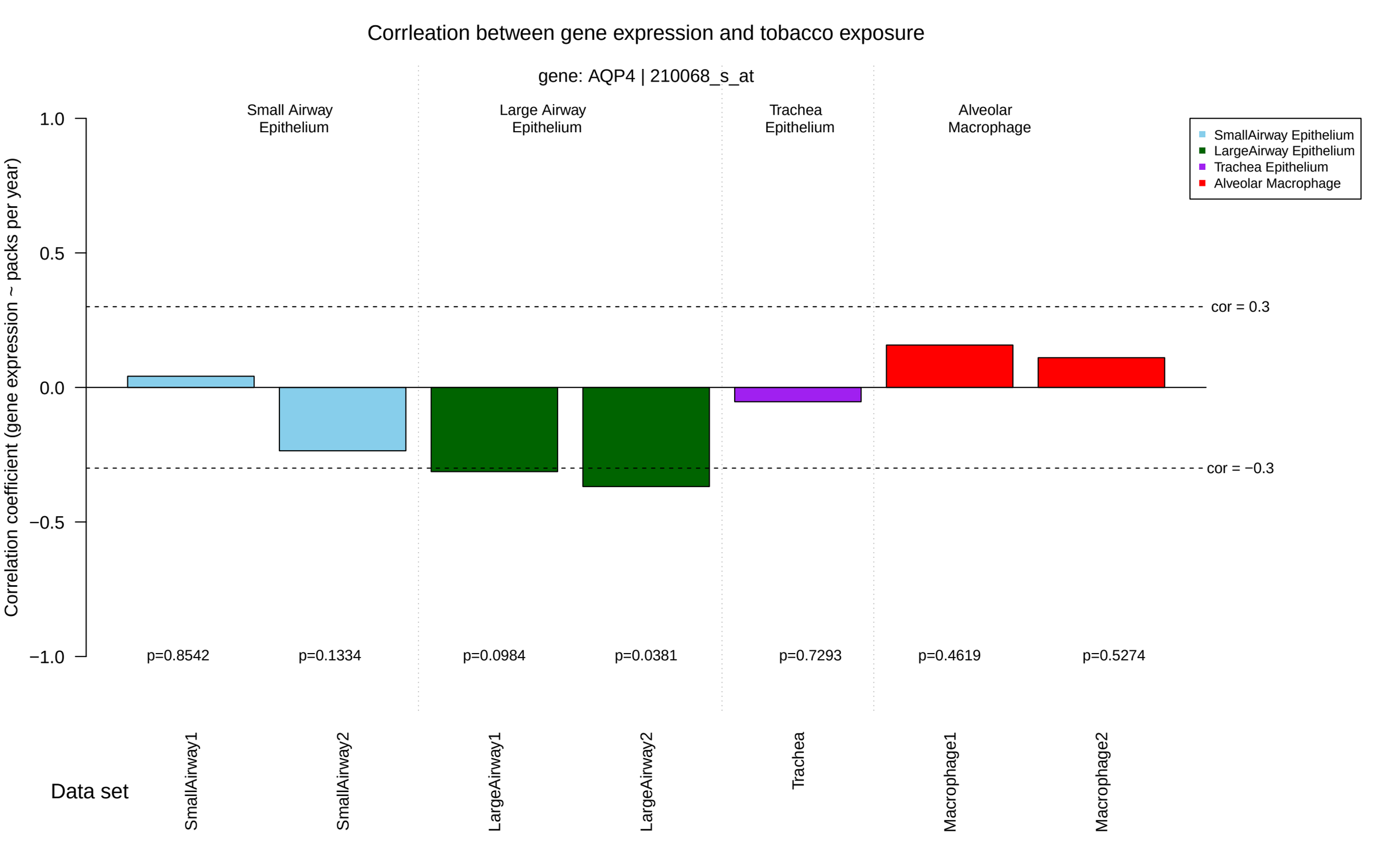

set: LargeAirway1 (p=0.0452)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0063)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.293)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6902)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2065)
AQP4 | 210068_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP4 | 210068_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8542
p=0.1334
p=0.0984
p=0.0381
p=0.7293
p=0.4619
p=0.5274
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2